Machine Learning - Mini Project III

Matin Moradi - Abo Akademi University

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1. Introduction

Moisture content is really important when it comes to using biomass for energy production. If the biomass has much moisture it reduces the amount of energy we can get from it and increases the cost of transporting it and makes it harder to burn properly. The usual way to measure moisture content is by drying the biomass in an oven. This takes a long time. And we cannot do it while the biomass is moving.

There is a way to measure moisture content and that is by using near-infrared (NIR) spectroscopy and machine learning. This method is quick. Does not damage the biomass. When we shine NIR light on the biomass the water molecules absorb some of the light at specific wavelengths (4 000–12 000 cm^{-1}). The amount of light absorbed tells us how much moisture is in the biomass. To get a prediction of the moisture content we need to follow a few steps: preprocessing, optional dimensionality reduction, and a regression model. This project looks at ways of doing these steps using a set of data from an industrial biomass dataset.

2. Data Analysis

Dataset. Spectra were collected from biomass blends. They used a spectrometer with a contactless head at a belt speed of 1 meter per second. Each observation x is a feature vector of size 1037 and the target y is the moisture content, in percent.

Table 1 summarizes the parameters. The distribution of moisture is shown in Figure 1.

Table 1: Dataset summary.

Spectral range	4 000–12 000 cm^{-1} (834–2500 nm)
Points per scan	1 037
Total scans	773 (5–7 replicates/sample)
Physical samples	126 (after replicate averaging)
Moisture range	18.0–73.1% (mean 42.5%, std 14.4%)

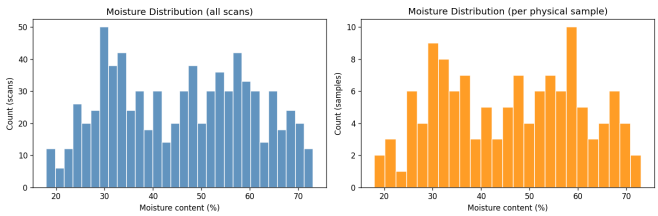

Figure 1: Moisture distribution: (a) all 773 scans, (b) 126 averaged samples.

Dataset challenges. (1) *High dimensionality* ($p = 1037 \gg n = 126$): The dataset has a lot of features which is a problem. We have 1037 features and just 126 samples.

(2) *Replicate leakage*: Another challenge is that we have multiple scans per sample. If we just randomly split these repeated scans into groups we will get error estimates that are too optimistic. To avoid this we need to use methods like grouped cross-validation or average the results of the repeated scans.

3. Pre-processing

First, 39 scans (5%) were flagged as outlier by PCA-based Hotelling T^2 and Q-residual statistics. After that, I averaged the duplicate scans per sample and then, we need to apply a preprocessing technique.

Raw NIR spectra of biomass are dominated by changes in the baseline and differences in how the light scatters. These changes can be larger than the signal that is related to moisture. Without correction, regression models learn physical artefacts rather than chemistry. So we need to pre-process the data to get rid of these variations that are not important while still keeping the parts of the spectra that are related to moisture in the biomass.

Eight pre-processing methods were used. Multiplicative Scatter Correction (MSC), Standard Normal Variate (SNV), Savitzky–Golay 1st &2nd derivative (SG1 & SG2), and their combinations. They were evaluated using 10-fold grouped CV RMSE with PLSR (latent variables tuned per method) on the 126 averaged samples. Results are in Table 2 and Figure 2.

MSC achieves the lowest RMSECV (2.202%, LV = 10) and was selected as the final pre-processing method. SNV is a practically equivalent alternative (2.222%) while SG1 and SG2 performing worst. Figure 3 shows raw vs. MSC-corrected spectra.

Table 2: Pre-processing ranked by grouped CV RMSECV (PLSR, 10-fold, n=126).

Method	RMSECV (%)	Best LV
MSC	**2.202**	**10**
SNV	2.222	12
SNV+SG1	2.376	8
SNV+SG2	2.643	6
Raw	2.666	14
Detrend	2.744	12
SG1	2.980	9
SG2	3.043	7

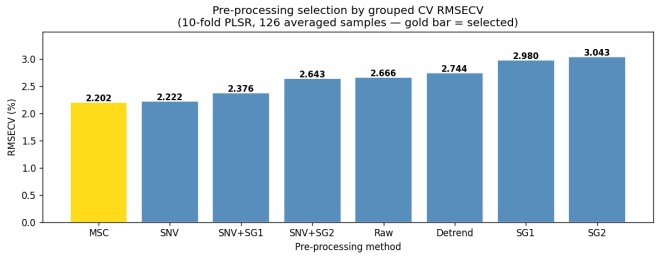

Figure 2: Pre-processing selection by grouped CV RMSECV. Gold bar = MSC (selected).

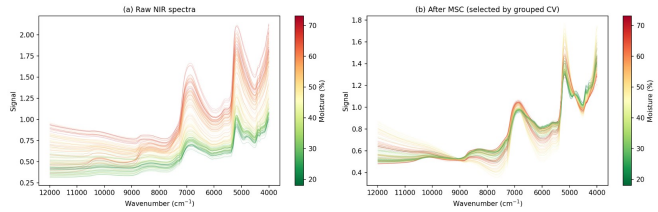

Figure 3: 150 scans coloured by moisture (green=low, red=high). (a) Raw: scattering dominates. (b) MSC: baseline removed.

4. Modelling Approaches

Before any modelling, the 126 averaged samples were split into two stratified holdouts, 80% for training and 20% for test. Stratification on 5 moisture quantile bins ensured both sets cover the full moisture range, giving 100 training and 26 test samples. The test set was locked and never used for fitting, scaling, PCA, or hyperparameter tuning.

Within the training set, all hyperparameters were tuned by 10-fold `GroupKFold` cross-validation (each averaged sample is its own group).

About the performance metrics, RMSECV was used on the training set and RMSE, R^2, and RPD were used on the test set. However, I mainly focused on these two:

- **RMSECV** — Root Mean Squared Error from 10-fold CV on the training set ($n = 100$). Used for hyperparameter selection.
- **RMSE** — Root Mean Squared Error on the independent 26-sample test set, after the final model is re-fitted on all 100 training samples.

Five models were evaluated:

PLSR Partial Least Squares regression decomposes both **x** and **y** into a set of shared latent variables, maximizing the covariance between them. The number of LVs was tuned by 10-fold CV on the training set (range 1–25) and optimal was LV = 13. Figures 4 and 5 show the tuning curve and the regression coefficients.

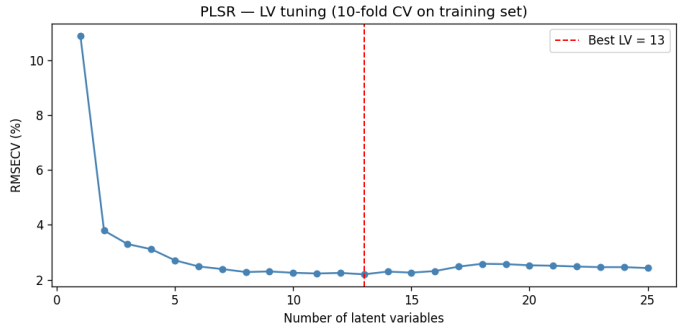

Figure 4: PLSR latent variable tuning

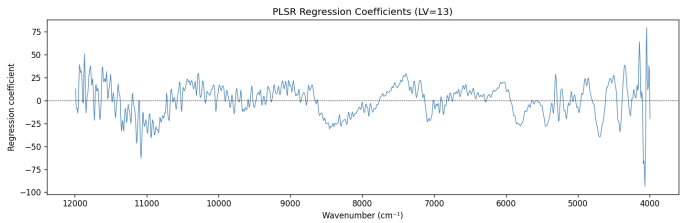

Figure 5: PLSR regression coefficients

SVR-PCA: I applied PCA to get 6 most important components (99.9% variance) and then applied SVR with RBF kernel. Optimal hyperparameters from grid search: $C = 100$, $\varepsilon = 0.01$.

SVR-Full: same SVR but this time on all 1 037 wavelengths. Optimal parameters: $C = 300$, $\varepsilon = 0.3$

ANN-PCA: MLP trained on 6 PCA components. Architecture: 1 hidden layer (optimal = 10 neurons), ReLU, L2 ($\alpha = 0.01$), `lbfgs` solver.

ANN-Full: Same MLP (hidden = 30, ReLU, $\alpha = 0.01$, `adam` solver) but this time on all 1 037 wavelengths.

5. Results

Table 3 reports all metrics for the five models, sorted by test RMSE. Parity plots, RMSECV vs. RMSE, and model rankings are shown in Figures 6–8.

Table 3: Five-model comparison (MSC pre-processing)

Model	RMSECV (%)	RMSE (%)	R^2	RPD
SVR-Full	2.589	**2.050**	**0.980**	**7.14**
SVR-PCA	3.339	2.260	0.976	6.48
PLSR	**2.195**	2.388	0.973	6.13
ANN-PCA	4.911	3.054	0.957	4.79
ANN-Full	11.39	9.954	0.538	1.47

SVR-Full achieves the best test performance (RMSE = 2.05%, R^2 = 0.980, RPD = 7.14) despite its RMSECV (2.59%) being higher than PLSR's (2.20%). This gap is related to an interesting concept named Cross-validation (CV) pessimism which means When you evaluate a model using cross-validation, the estimated error can be slightly higher than the true generalization error. Why it happens? Because the model is trained on a smaller dataset in the CV phase and when you train it on the whole training dataset, the performance improves. This can especially happen in our setup

because the size of our training dataset (100) is small itself and its dimensionality is high, so when we use kfold cross validation even smaller data (90) will be left for an accurate model training.

Another point is that feeding the full 1 037-wavelength MSC spectrum provides richer geometric information than PCA-compressed inputs. This is why SVR-Full is performing better than SVR-PCR.

SVR-PCA ranks second (RMSE = 2.26%, RPD = 6.48). Like SVR-Full, it benefits from the non-linear kernel. Its RMSE is lower than RMSECV which confirms the pattern of CV pessimism for support vector methods at this sample size.

PLSR gives the lowest RMSECV (2.20%), but it ranks third on the test set (RMSE = 2.39%). Since PLSR is a linear method, its parameters do not change much when some samples are left out during cross-validation. This makes its CV results stable and reliable. However, SVR uses a non-linear kernel, which can capture more complex relationships between moisture and spectral data. This advantage becomes clear only when the model is trained on the full dataset.

ANN-PCA achieves an RPD = 4.79, which is considered a good result. The lbfgs solver converges reliably when using 6 PCA components with L2 regularization ($\alpha = 0.01$). However, its performance is still lower than that of the kernel-based methods.

ANN-Full performs poorly (RPD = 1.47), which is not acceptable. The model has about $1037 \times 30 = 31\,110$ parameters but is trained on only 100 samples, making it heavily over-parameterized (about 311 times more parameters than data). As a result, it fails to learn meaningful patterns and just predicts values close to the training mean.

Figure 6: Parity plots on the test set (n=25) across the full 18–73% moisture range.

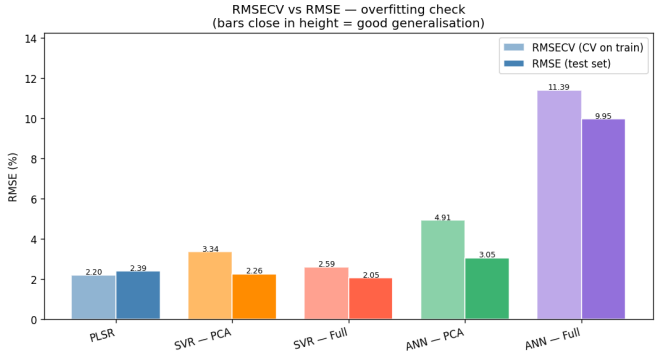

Figure 7: RMSECV vs. test RMSE

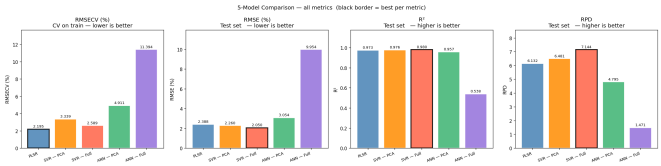

Figure 8: Comparing all 5 models across all 4 metrics

6. Conclusions

Results achieved. In this work I looked at five machine learning models to see how well they could predict moisture content in biomass. These machine learning models were trained and tested using 126 averaged FT-NIR biomass spectra. All top four models achieve RPD > 4, and three of them with the RPD > 5 excellent threshold.

Best ML approach. SVR with the full spectrum (SVR-Full) achieves the highest accuracy on the independent test set. Its RBF kernel can capture complex and nonlinear relationships between moisture and spectral data without greatly increasing the number of parameters. It uses all of the 1,037 columns of spectra data which preserves more information compared to PCA-compressed inputs and gives better results.

PLSR performs best in cross-validation (lowest RMSECV) and is the most stable model, but because it is linear, its accuracy is limited. In contrast, SVR can achieve better performance when trained on the full dataset.

Best pre-processing. MSC was the best choice based on cross-validation (RMSECV). Using scatter correction alone (MSC: 2.202%, SNV: 2.222%) performed better than all combinations with Savitzky–Golay derivatives on this small dataset (126 samples). The thing is derivatives can make the high-frequency noise worse. When we do not have a lot of data this noise hurts our results more, than the good that comes from removing baseline effects.

Improvements There are two types of dimensionality reduction: feature selection and feature extraction. The first one Keeps good and original variables and removes unimportant ones such as GA (Genetic Algorithm) and iPLS (interval PLS). On the other hand, feature extraction methods creates new variables (latent features) and somehow combine original variables like PCA, LDA, and PLS. Here I just used PCA. One improvement can be using feature selection methods to select just the important wavelengths (as you can see in Figure 3 different intervals in the spectra don't have the same level of effect).

Bottlenecks. The main limitation is the small dataset (n = 126) with a high dimensionality. This makes it hard to train complex models like ANN, increases variability in cross-validation results, and limits reliable comparison between similar models. A larger and more diverse dataset would address most of these issues.